STAFF REVIEW · H446 101 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A CPU processing unit able to run a hardware thread or more, depending on design.

2. Several tasks progress over overlapping time.

3. Serial work limits parallel speedup.

4. 12.

I Do — source worked steps

1. Explain the choice — 8 ÷ 2 = 4 equal-duration waves, versus 8 on one core.

We Do — guided CHECK

1. Explain the choice
12 ÷ 4 = 3 waves; shared-memory contention or scheduling overhead can extend time.

You Do — source mark schemes

1. Three cores 12÷3=4 waves [3], one core12 waves [2], ideal threefold throughput gain [1].

2. Divide independent frames/tiles [2], throughput [2], serial dependencies [2], contention/overhead [2].

3. Normalised time 0.5+0.5/4=0.625 [3], speedup 1/0.625=1.6 [2], not fourfold due serial work [1].

Exit Ticket — source answer

1. False. Dependencies and overhead reduce gains.